



AI Trading Assistant

Intelligent Intraday Signals | AI Chat | Portfolio Management

Built on AWS Lambda + Bedrock + Streamlit

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Why I Built This



Signal Noise

Manual scanning of 200+ stocks daily was slow and unreliable — needed automated intelligence



Missed Entries

By the time patterns were spotted, the optimal entry window had often closed



AI-Powered Edge

Combine 12 technical indicators with Bedrock AI validation for higher-confidence signals



Risk Discipline

Built-in R:R ratio enforcement — never enter a trade without knowing the risk/reward upfront

Key Features



Signal Engine

12 indicators: RSI, MACD, BB, ATR, OBV, Stoch, ADX, MFI, VWAP, EMA, SMA, RS



AI Validation

Claude Haiku validates every signal — confirms or rejects with reasoning & confidence score



Index Strip

NIFTY 50, BANKNIFTY, VIX always visible — live prices via direct Yahoo Finance v8 API



Portfolio Tools

Personal watchlist, trade log, performance metrics — all isolated per Cognito user



Real-time Data

AWS Lambda pulls live quotes, processes signals, stores results in DynamoDB



R:R Badges

Every stock card shows Risk:Reward ratio — colour-coded green/amber/red for quick decisions

Signal Algorithm — 12 Indicators

RSI	Overbought/oversold momentum	MACD	Trend momentum crossover
Bollinger Bands	Volatility squeeze signals	ATR	True range for stop-loss sizing
OBV	Volume confirms price moves	Stochastic	K/D oscillator divergence
ADX	Trend strength filter (>25)	MFI	Money Flow Index
VWAP	Volume-weighted anchor	EMA 9/21	Short-term trend direction
SMA 50/200	Long-term trend regime	RS Ratio	$(1+stock)/(1+nifty)$ vs benchmark

Scoring Logic

8-12 pts	STRONG BUY
5-7 pts	BUY
3-4 pts	HOLD
1-2 pts	SELL
0 pts	STRONG SELL

+ AI Validation (Bedrock Claude Haiku)
confidence score added to final rank

Architecture



Frontend

Streamlit UI

Index Strip
(NIFTY/BANKNIFTY/VIX)

Signal Dashboard

AI Chat

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AWS Lambda

Signal Engine (12 indicators)

AI Validator (Bedrock Haiku)

Sentiment Agent (Nova Micro)

Yahoo Finance v8 API



Data Layer

DynamoDB — signals &
portfolio

S3 — historical data cache

Cognito — JWT auth, per-user
isolation

EventBridge — scheduled
triggers



Auth & Security:

Cognito JWT tokens — per-user DynamoDB isolation. Email verification required. All Lambda calls are signed AWS API requests.

AI Chat — Ask the Market

You:

Why is RELIANCE showing a S

AI:

RELIANCE is up 2.3% on OBV surge with RSI at 58 (momentum). MACD crossed bullish 3 bars ago. RS Ratio vs NIFTY is 1.04 — outperforming. Entry: 2847, Target: 2940, Stop: 2800.

You:

What's the market sentiment

AI:

NIFTY +0.8% | BANKNIFTY +1.1% | VIX 13.2 (calm). 68% of scanned stocks show bullish bias. Sector leaders: IT, Banking.

Capabilities



Signal explanations with indicator breakdown



Market context using index strip data



Portfolio analysis & trade reviews



Risk assessment per position



Bedrock Claude 3.5 Sonnet backbone



Cognito-isolated chat history per user

Portfolio & Wishlist



Portfolio Tracker



Log trades with entry price, quantity, date



Live P&L calculated against current price



Per-trade return % and absolute gain/loss



CSV export for tax/reconciliation



All data isolated per Cognito user



Wishlist (Watchlist)



Add any NSE stock to personal watchlist



Live signal updates every 5 minutes



Quick-access from main dashboard



Wishlist signals trigger push summary



Per-user isolation via JWT

Tech Stack

Frontend

Streamlit

Python 3.11

Plotly

Boto3 SDK

Compute

AWS Lambda

Python 3.11

yfinance / Yahoo v8 API

pandas / numpy

AI / ML

Bedrock Claude 3.5 Sonnet

Claude Haiku (validator)

Nova Micro (sentiment)

Prompt templates

Data & Auth

DynamoDB

S3

Cognito (JWT)

EventBridge cron

Infra / CI

Terraform (IaC)

GitHub Actions

Lambda layers

CloudWatch logs

Cost & Scalability

~\$3

per month
AWS cost

<3s

Lambda cold
start p95

200+

stocks scanned
per cycle

5 min

signal refresh
cadence



Serverless by design

Lambda + DynamoDB auto-scales — no idle EC2 costs.
Pay only per invocation.



DynamoDB on-demand

No capacity planning. Handles 1 user or 1,000 users
with same infrastructure.



Scheduled scans

EventBridge cron triggers Lambda every 5 min during
market hours (9:15–15:30 IST).



Cost ceiling

Free tier covers most usage. AWS Budgets alert set at
\$10/month threshold.

Live Market Intelligence Strip

NIFTY 50

24,531.50

+0.82%

BANKNIFTY

52,140.30

+1.14%

INDIA VIX

13.24

+2.1%

5 min

refresh



Always Visible

Index strip is pinned in the Streamlit header — visible on every page of the app without scrolling



Sub-3s Response

Direct Yahoo Finance v8 API calls from Lambda — no yfinance library overhead that caused timeouts



Market-Aware Display

Shows last close price when market is closed. Switches to live price automatically during 9:15–15:30 IST trading hours



Colour-Coded Moves

% change shown in green (up) or red (down) — instant visual context without reading the numbers

Signal Dashboard — At a Glance



Top Movers



Signal Summary Cards

One-line snapshot of STRONG BUY / BUY / HOLD / SELL / STRONG SELL counts + Avg AI Score at top of dashboard



Top Movers Strip

Scrolling pills showing biggest % movers always pinned above the signal grid — spot leaders instantly



R:R Ratio Badge

Every stock card shows Risk:Reward ratio. Green badge $\geq 2:1$ | Amber $\geq 1:1$ | Red $< 1:1$



Signal Timestamps

Each signal shows when it was last computed (IST) — know if you're looking at fresh or stale data

Signal Quality Fixes

BUG FIX 1

Target Price for BUY Signals



Problem:

Breakout stocks were showing target price below current price — indicating a sell when signal was BUY

Fix:

Target is now always capped to $\text{current_price} * 1.02$ minimum for BUY signals, ensuring $\text{target} > \text{entry}$

Impact:

No more contradictory signal/target pairs confusing trade decisions

BUG FIX 2

RS Ratio Formula Correction



Problem:

Old formula: $\text{stock_return} / \text{nifty_return}$ — inverted when NIFTY was negative, making weak stocks appear strong

Fix:

Corrected to $(1 + \text{stock_return}) / (1 + \text{nifty_return})$ — handles negative benchmark correctly

Impact:

Relative strength now accurately identifies outperformers vs the index in all market conditions

BUG FIX 3

Support / Resistance Selection



Problem:

Silent None bug: when pivot levels were empty, nearest support/resistance returned None — causing downstream crash

Fix:

Added None guard with fallback to $\text{current price} \pm \text{ATR}$ when no pivot levels are available

Impact:

Stop-loss and target calculations now always return valid prices — no more silent failures



All 3 fixes ship in the same release — more accurate signals = better trade decisions